1. (a) $\because |Z_n^*| \le n-1$ 当n为素数取等.

$\therefore \frac{|Z_n^*|}{2} \le \frac{n-1}{2}$

$\because G(n)$是Z_n^*的一个子群

由拉格朗日定理 $|G(n)| \mid |Z_n^*|$

$\therefore |G(n)| \le |Z_n^*|$

又$\because G(n) \ne Z_n^*$

$\therefore |G(n)| < |Z_n^*|$

又$\because |G(n)| \mid |Z_n^*|$

$\therefore |G(n)| \le \frac{|Z_n^*|}{2}$

综上 $|G(n)| \le \frac{|Z_n^*|}{2} \le \frac{n-1}{2}$

(b) 由二项展开 $(a+b)^m = \sum_{i=0}^{m} C_m^i a^{m-i} b^i$

$\therefore (1+p^{k-1}q_k)^{\frac{n-1}{2}} \equiv 1 + \frac{n-1}{2} p^{k-1} q_k \bmod n$

又$\because \left(\frac{1+p^{k-1}q_k}{n}\right) = \left(\frac{1+p^{k-1}q_k}{p^k}\right)\left(\frac{1+p^{k-1}q_k}{q_k}\right)$

$\because 1+p^{k-1}q_k \equiv 1 \bmod p$

$\therefore \left(\frac{1+p^{k-1}q_k}{p^k}\right) = \left(\frac{1+p^{k-1}q_k}{p}\right)^k = 1$

又$\because 1+p^{k-1}q_k \equiv 1 \bmod q_k$

$\therefore \left(\frac{1+p^{k-1}q_k}{n}\right) = 1$

$\because \frac{n-1}{2} p^{k-1} q_k \not\equiv 0 \bmod n$

$\therefore \left(\frac{a}{n}\right) \not\equiv a^{\frac{n-1}{2}} \pmod n$

(c) $\left(\frac{a}{n}\right) = \prod_{i=1}^{s} \left(\frac{a}{p_i}\right)$

$\because a \equiv 1 \bmod p_2 p_3 \cdots p_s$

且 $a \equiv u \bmod p_1$, u是p_1的二次非剩余

$\therefore \left(\frac{a}{n}\right) = (-1) \cdot 1^{s-1} = -1$

$\because a \equiv 1 \bmod p_2 \cdots p_s$

$\therefore a^{\frac{n-1}{2}} \equiv 1 \bmod p_2 \cdots p_s$

若 $a^{\frac{n-1}{2}} \equiv -1 \bmod n$

则 $a^{\frac{n-1}{2}} \equiv -1 \bmod p_2 \cdots p_s$

矛盾

$\therefore a^{\frac{n-1}{2}} \not\equiv -1 \bmod n$

(d) 由 b, c 可知, $\left(\frac{a}{n}\right)$不满足欧拉判别法

$\therefore a \notin Z_n^*$

$\therefore G(n) < Z_n^*$

又$\because |G(n)| \mid |Z_n^*|$

$\therefore G(n) \le \frac{n-1}{2}$

(e) $\because G(n) \le \frac{n-1}{2}$

$\therefore$ 只有取到 $G(n)$中的元素, 才会输出是素数

$\therefore P = \frac{|G(n)|}{|Z_n^*|} \le \frac{1}{2}$

$\therefore$ 至多为$\frac{1}{2}$